

SentinelQA: Task-Level Routing and Output Verification for Multilingual Instruction Following

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Abstract

We describe team Sentinel’s submission to the WMT26 MIST shared task on multilingual instruction following. We allocate the budget across a pipeline built on `gemma-3-4b-it`: a fixed task-level router that selects between fine-tuned and base behaviour, and a deterministic verification layer built from language identification, a length parser and a repetition detector. The LoRA adapter can be served as an adapter toggled against one shared copy of the base weights, giving a deployed footprint of 5.16B rather than the **9.46B** our submitted outputs used with a merged fine-tune. Fine-tuning raises context-based QA token-F₁ from 0.267 to 0.74. The verification layer alters roughly 9% of outputs and raises our reference-free compliance measure by 2.2 points; because a replacement is accepted only when it passes more of the same checks, we report this as the yield of the intervention rather than an identified effect.

1 Introduction

The WMT26 MIST shared task evaluates multilingual instruction following across three subtasks: context-based question answering (`qa-context`), open-ended generation QA (`qa-oeg`), and mono- and cross-lingual summarization (`sum-sum`), in 24 languages and subject to a limit of 10B parameters across all deployed components.

The test set contains 12,775 items: 8,640 `qa-context` (67.6%), 2,359 `qa-oeg` (18.5%) and 1,776 `sum-sum` (13.9%). Most items are cross-lingual. Computed from the item identifiers, 95.8% of `sum-sum` items and 96.1% of `qa-context` items ask for output in a language other than that of the source document, almost always English; `qa-oeg` is entirely monolingual.

Our submission is built on `gemma-3-4b-it` (Gemma Team, Google DeepMind, 2025), a 4.37B-parameter instruction-tuned model. We submit three systems. The primary system routes each

item by subtask: fine-tuned behaviour for `qa-context`, base behaviour for `qa-oeg` and `sum-sum`. The two contrastive systems are the fine-tuned model alone and the base model alone, the latter doubling as our untuned baseline. All three pass every generation through the same verification layer, shown in Figure 1. (§4).

2 Related Work

The verification layer is a minimal instance of the pattern studied in the self-correction literature, e.g. Self-Refine (Madaan et al., 2023) and constraint decomposition approaches such as DECRIM (Ferraz et al., 2024), but its checks are external and non-neural rather than model-judged. Huang et al. (2024) report that LLMs struggle most with autonomous error *detection*, which is the step we delegate to fastText classifiers (Joulin et al., 2017, 2016) trained for language identification (Kargaran et al., 2023; Burchell et al., 2023) and to deterministic parsers. In machine translation, reranking an n -best list with a quality-estimation model (Rei et al., 2020; Fernandes et al., 2022; Freitag et al., 2022) is the established way to recover from a bad sample; our single-candidate retry is a cheaper point on that trade-off. Following the WMT25 MIST findings (Kocmi et al., 2025) we treat chrF++ (Popović, 2017) as reportable but not decisive, given its weak correlation with human judgment on low-resource languages.

3 Data

We train on the sample release provided by the shared-task organizers (Chen and WMT26 MIST Organizers, 2026), whose per-language table sums to the 23,919 rows we audited.

Its sources are TyDi QA (Clark et al., 2020), BELEBELE (Bandarkar et al., 2024), MCIF (Papi et al., 2025), the Aya dataset (Singh et al., 2024), CrossSum (Bhattacharjee et al., 2023), WikiLingua

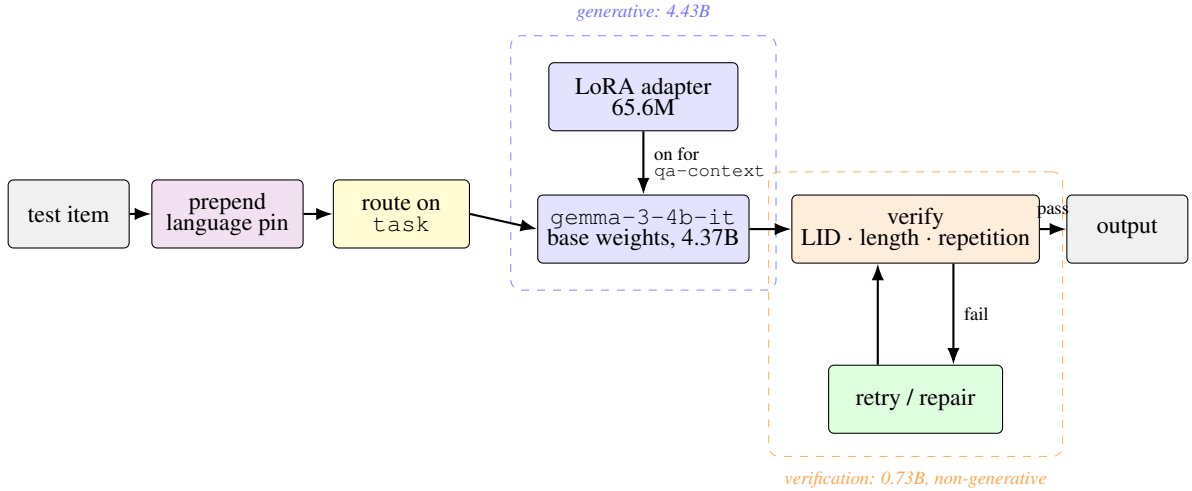


Figure 1: The SentinelQA pipeline. The router selects a *behaviour* rather than a model: the adapter is enabled for qa-context and disabled for qa-oeg and sum-sum, so one copy of the base weights serves both paths. Every generation is prefixed with a target-language pin and checked deterministically, and a retry or repair replaces the original only if it passes more checks than the original did. Our submitted outputs were produced with a merged full fine-tune in place of the adapter, which needs a second copy of the weights (§6.1).

(Ladhak et al., 2020) and the WMT25 MIST OEG submissions.

We first froze an evaluation split, sampling $\min(100, \lceil 0.15n \rceil)$ items from each (language, subtask) cell into a held-out set of 3,599 rows, then made it read-only and hash-pinned. For cells with no data at all — Yoruba qa-context, for instance — we filled with teacher distillation for qa-oeg (2,043 rows), NLLB-200-3.3B (NLLB Team et al., 2022) translation into six languages (747 rows) and native BELEBELE qa-context for Bengali, Czech and Yoruba (264 rows). Augmented rows passed the same language and degeneracy checks used at inference. The final corpus is 23,117 rows.

Two properties of the data bear on the numbers below. BELEBELE is parallel, so one passage appears in every language under a shared identifier; we recovered identifiers for all 847 held-out BELEBELE rows and excluded from training every row carrying one of the resulting 400 identifiers, which removes 84% of otherwise-usable BELEBELE but keeps held-out passages out of training. Separately, 48 held-out rows carry an English reference despite a non-English target, all from the three languages outside WikiLingua’s 18-language coverage; we score them as a separate slice, so held-out results below use the remaining $n=3,551$.

4 System

Routing. The router is a fixed lookup on the task field with no learned component: qa-context takes fine-tuned behaviour, qa-oeg and sum-sum take base behaviour. Items are copied verbatim from the corresponding stream, so the routed submission is a deterministic recombination of two systems we also submit separately.

The three subtasks rest on evidence of different strength. qa-context (67.6% of the test set) is decided by a large margin: token- F_1 rises from 0.267 to 0.74 (§6.1). sum-sum (13.9%) is decided by length: the fine-tuned models produce a median of 92–134 words on prompts requesting roughly 200, where base produces 210. qa-oeg (18.5%) has neither. Its post-verification compliance differs by only 0.0025 (0.9806 base against 0.9781 fine-tuned) and we have no reference-based accuracy measure for open-ended generation, so we routed it to base for consistency with sum-sum, both being free-form generation subject to the same shortening. This is the weakest of the three decisions, assigning nearly a fifth of the test set on a tiebreak and an argument by analogy.

Verification. Every generation is prefixed with a target-language system pin (“Respond only in *<endonym>*”) and checked on three axes.

Language uses dual fastText identification with GlotLID v3 and OpenLID v3, returning PASS/FAIL/UNSCORABLE and rejecting text only

when *both* models place it outside the accept set, with short strings marked unscorable rather than failed. **Length** uses a deterministic multilingual parser for explicit word- and sentence-count constraints with script-aware counting; it fires on 85.6% of test prompts, uniformly across languages, covering 100% of `qa-context` and `sum-sum` prompts and 22.1% of `qa-og` prompts. **Degeneracy** uses n -gram repetition detection.

We summarise these as a **compliance** score: per item, the mean of the applicable binary checks — in the target language, non-degenerate, free of chat-template artefacts, and within a stated length constraint where one is parsed (Appendix A). Compliance is reference-free, so it can be computed on the test set, and it measures whether an output is well-formed rather than whether it is correct.

A failed check triggers a temperature-0 retry with a fixed correction appended; outputs still degenerate afterwards are regenerated with a repetition penalty, and under-length `sum-sum` drafts get one edit-based expansion pass. In every case the candidate replaces the original only if it passes strictly more checks, or fixes the language without breaking another axis.

Parameter budget. The router needs two *behaviours*, not two sets of weights. Our submitted outputs were generated with a merged full fine-tune, which requires a second complete copy of the model and totals **9.46B** once the language identification models are counted (Table 1, left). But Run C shows that a LoRA adapter of 65.6M parameters is statistically indistinguishable from that full fine-tune (§6.1), and an adapter need not be merged: current serving stacks apply LoRA weights at inference against a shared base, so the same routing reduces to enabling the adapter for `qa-context` and disabling it otherwise. That configuration holds one copy of the base weights and totals **5.16B** (Table 1, right), a 45% reduction for a system we cannot distinguish from the one we submitted.

We count the fastText models throughout, since they run at inference and are not negligible at 0.73B; under the adapter accounting they are the second-largest component after the base model itself.

5 Experimental Setup

All training used a single NVIDIA A100-SXM4-80GB with `bfloat16`, sequence length 4096 with packing, loss on assistant tokens only, effec-

component	as submitted	adapter
<code>gemma-3-4b-it</code> (base)	4,365,656,432	4,365,656,432
second copy, merged FT	4,365,656,432	—
LoRA adapter	—	65,576,960
GlottLID v3	418,934,528	418,934,528
OpenLID v3	307,181,056	307,181,056
total deployed	9,457,428,448	5,157,348,976

Table 1: Deployed parameters against the task’s 10B limit. The submitted outputs use the merged configuration; the adapter configuration serves the same two behaviors from one copy of the base weights, and is available because Run C is equivalent to the full fine-tune (§6.1). We did not regenerate the submission under it.

	Run A	Run B	Run C
data	sample	full	full
rows	20,063	23,117	23,117
method	full FT	full FT	LoRA
learning rate	1e-5	1e-5	1e-4
steps	141	159	159
final train loss	—	1.759	1.784

Table 2: The three training runs, which read the same corpus file and differ only in the source filter and tuning method. Run A’s final loss was not recorded.

tive batch size 64, three epochs, a cosine schedule with 3% warmup and seed 20260801. We trained three models (Table 2): Run A on the organizer sample only, Run B on the full augmented corpus, and Run C on the full corpus with LoRA (Hu et al., 2022) ($r=32$, $\alpha=64$, all attention and MLP projections, 65.6M trainable parameters, 1.50% of the model). Run C’s adapters are merged into the base weights at each checkpoint save for evaluation convenience, so all three runs are interchangeable under the same inference code; §4 discusses serving the adapter unmerged for deployment. Inference used vLLM (Kwon et al., 2023).

Checkpoints were selected by a rule fixed before any results existed: compliance plus `qa-context` F_1 on the native held-out slice, subject to an English/German retention gate, with chrF++ reported but not decisive (computed with sacreBLEU, Post, 2018). Both full fine-tunes peaked at epoch 2.

Comparisons use paired bootstrap resampling (Koehn, 2004) with $B=1000$; the finest p -value this resolves is $1/B$.

system	compl.	F ₁	chrF++	agg.
base	0.8073	0.2671	25.09	1.0744
Run A (e2)	0.8266	0.7462	45.17	1.5728
Run B (e2)	0.8238	0.7405	44.98	1.5643
Run C (e3)	0.8154	0.7499	45.39	1.5653

Table 3: Held-out results for the selected checkpoint of each run. “agg.” is the unweighted sum of compliance and qa-context F₁; it has no independent interpretation and is reported only because it was the pre-registered selection rule.

6 Results

Held-out numbers use the native slice ($n=3,551$; F₁ over the $n=1,454$ qa-context items).

6.1 Held-out results

Table 3 gives the selected checkpoint of each run against the base model. Fine-tuning raises qa-context token-F₁ from 0.267 to roughly 0.74 across all three runs. Since the gold spans have a median of four words while the base model emits a median of 30 on the same items, much of this margin reflects acquisition of the expected output format rather than improved comprehension. Compliance moves little, and the base model is marginally ahead of Run B on it. Appendix B gives a per-language breakdown.

Against the base model, Run B gains +0.4743 F₁ (95% CI [+0.4521, +0.4976], $p < 1/B$).¹ Differences *among* the three fine-tuned systems are best read as equivalences: all pairwise compliance intervals fall within ± 0.004 and all F₁ intervals within ± 0.021 , so full fine-tuning, LoRA and our two data compositions are indistinguishable at this sample size, and the ordering in Table 3 is not a ranking.

6.2 Verification layer

Our artefacts record the raw first generation alongside the shipped text, so the layer can be evaluated on the full test set without regenerating anything (Table 4). The three single-model systems alter 8.6–9.9% of outputs and add 2.4–2.8 compliance points; the routed system, whose items are copied from those streams, adds 2.2. Because the acceptance rule of §4 keeps a replacement only when it passes more of the same checks that compliance measures, these figures are the yield of the interven-

¹Computed on Run B’s epoch-3 checkpoint; the submitted epoch-2 checkpoint differs from base by +0.4734 on the same measure.

system	altered	raw	shipped	yield
Run B alone	9.9%	0.9436	0.9715	+0.0279
Run A alone	9.2%	0.9374	0.9626	+0.0252
base alone	8.6%	0.9512	0.9749	+0.0237
routed	—	0.9554	0.9773	+0.0219

Table 4: Verification layer over all 12,775 test items. “altered” is the share of outputs that did not ship as first generated. The routed row is an item-weighted combination of the three streams.

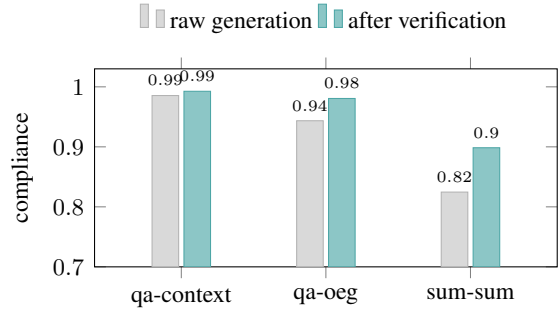


Figure 2: Compliance before and after verification for the routed system, following each subtask to the model that serves it: +0.007 on qa-context (from Run B), +0.037 on qa-oeg and +0.074 on sum-sum (both from base).

tion rather than an independently identified effect, and we report them without confidence intervals for that reason.

Figure 2 shows where the yield arises in the routed system: little on qa-context, which is already near ceiling, and more on the two generation subtasks, where raw output is weakest.

6.3 Ablations

LoRA. Run C matches the full fine-tunes on both metrics within the equivalence bounds above, at 1.50% of the trainable parameters. It also matches their sum-sum output length (24 words held-out against 25 for Run A and Run B, versus 185 for base), so the shortening that motivates our routing rule is attributable to the training data — whose median sum-sum target is about 25 words — rather than to the tuning method.

Language pin. A 288-item probe over both models shows the English pin helps despite being written in a different language from the prompt: compliance is 89.2% with it versus 86.9% without for Run B, and 90.0% versus 85.0% for base, concentrated in sum-sum where base compliance falls to 45.8% without it. This is a small-scale replication of the language-confusion effect documented

by Marchisio et al. (2024), who find that models frequently generate in the wrong language and that explicit instruction mitigates it.

Augmentation. Run A and Run B differ only by the source filter, and are equivalent within ± 0.0024 compliance and ± 0.014 F_1 despite Run B carrying 3,054 additional teacher-distilled, machine-translated and native-BELEBELE rows. The per-language view (Appendix B) sharpens this into a negative result. Yoruba has no organizer qa-context rows at all, so that share of its data is entirely material we added, and it shows the largest regression of any language: 0.816 compliance under base against 0.675 under Run B. The augmented run is worse than the untuned model on a language the augmentation was specifically for.

7 Limitations

Compliance measures whether outputs are well-formed and not whether they are correct, and the verification figures in §6.2 are yields under a rule that optimises that same measure; establishing an improvement in quality would require a criterion we did not select on. We have no reference-based accuracy measure for qa-*oeg*, so the routing decision for that subtask rests on weaker evidence than the one for *sum-sum*. All runs used a single training seed, so the paired bootstrap bounds evaluation noise but not seed variance. Finally, our held-out set is drawn from the organizer sample and differs from the test distribution — most visibly in summary length and in how often prompts state an explicit length constraint (4.7% held-out against 85.6% of test prompts) — so held-out figures are best read as relative comparisons between our own systems rather than as predictions of test performance.

8 Conclusion

We submitted a routed system to WMT26 MIST: fine-tuned and base behaviour under a fixed task-level router, wrapped in a deterministic verification layer. Our submitted outputs used a merged fine-tune at 9.46B, and we did not regenerate them under the adapter configuration.

Fine-tuning improves context-based QA by a decent margin, substantially through output-format acquisition. The verification layer alters roughly 9% of outputs and adds 2.2 compliance points as a yield of its acceptance rule. Full fine-tuning, the

adapter and our two data compositions are equivalent within about half a point of compliance, which suggests that at this scale effort is better spent on how a system is served and on what it does at inference time than on the choice of tuning method.

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A Metric definitions

For an output y on item i , let $A(i)$ be the set of applicable checks: language (unless the target is unresolved or the text too short to score), degeneracy, chat-template artefacts, and length (only where a constraint is parsed). Compliance is the unweighted mean of the binary outcomes over $A(i)$:

$$\text{compl}(y_i) = \frac{1}{|A(i)|} \sum_{c \in A(i)} \mathbf{1}[c \text{ passes on } y_i],$$

and a system’s compliance is the mean over items. $|A(i)|$ is 4 where a length constraint is parsed and 3 otherwise, so the measure is not directly comparable between corpora with different constraint coverage. The selection aggregate in Table 3 is $\text{agg} = \text{compl} + F_1^{\text{qa-context}}$, an unweighted sum of two quantities on different scales, fixed in advance as a tie-breaking rule and reported only for transparency about how the submitted checkpoints were chosen.

B Per-language compliance

Table 5 gives compliance per language on the native held-out slice, aggregated over subtasks and weighted by item count. We report held-out rather than test figures because the per-item test artefacts were not retained; the held-out set covers 27 languages, all but Bhojpuri of the 24 test languages. This table should be read as indicative rather than as a ranking.

lang	n	base	B	C	lang	n	base	B	C
arb	214	0.916	0.909	0.845	mar	138	0.911	0.905	0.904
ben	77	0.747	0.750	0.750	pes	101	0.861	0.861	0.861
ces	7	1.000	1.000	0.964	por	161	0.881	0.901	0.906
ckb	45	1.000	1.000	1.000	rus	213	0.927	0.942	0.937
deu	152	0.850	0.977	0.967	slk	45	1.000	1.000	0.993
eng	157	0.970	0.959	0.941	spa	160	0.887	0.873	0.872
fin	45	0.744	0.750	0.750	swh	99	0.861	0.864	0.859
fra	158	0.888	0.876	0.878	tel	99	0.859	0.861	0.861
hat	45	0.985	0.993	0.993	tha	94	0.755	0.863	0.863
hin	168	0.808	0.838	0.857	tur	160	0.916	0.929	0.896
ind	213	0.878	0.873	0.872	vie	160	0.758	0.823	0.728
ita	153	0.994	0.975	0.957	yor	78	0.816	0.675	0.754
jpn	210	0.768	0.802	0.803	zho	199	0.802	0.857	0.865
kor	200	0.749	0.755	0.752	all	3551	0.807	0.824	0.815

Table 5: Compliance by language, native held-out slice, selected checkpoints. Run A is omitted: it is equivalent to Run B (§6.1) and tracks it to within 0.01 in every language except Yoruba, where it scores 0.761.